equations only provide the magnitude. The phase is determined solely by the left-right positioning of the time domain waveform, as discussed in the last chapter.

Notice in Fig. 11-3b that the amplitude of the oscillation does not decay to zero before a frequency of 0.5 is reached. As you should suspect, the waveform continues into the next period where it is *aliased*. This changes the shape of the frequency domain, an effect that is included in Eqs. 11-1 and 11-2.

It is often important to understand what the frequency spectrum looks like when aliasing *isn't* present. This is because *discrete signals* are often used to represent or model *continuous signals*, and continuous signals don't alias. To remove the aliasing in Eqs. 11-1 and 11-2, change the denominators from $sin(\pi kM/N)$ to $\pi kM/N$, and $sin(\pi f)$ to πf, respectively. Figure 11-4 shows the significance of this. The quantity πf can only run from 0 to 1.5708, since f can only run from 0 to 0.5. Over this range there isn't much difference between $sin(\pi f)$ and πf. At zero frequency they have the same value, and at a frequency of 0.5 there is only about a 36% difference. Without aliasing, the curve in Fig. 11-3b would show a slightly lower amplitude near the right side of the graph, and no change near the left side.

When the frequency spectrum of the rectangular pulse is *not* aliased (because the time domain signal is continuous, or because you are ignoring the aliasing), it is of the general form: $sin(x)/x$, i.e., a sinc function. For continuous signals, the *rectangular pulse* and the *sinc function* are Fourier transform pairs. For discrete signals this is only an approximation, with the error being due to aliasing.

The sinc function has an annoying problem at $x = 0$, where $sin(x)/x$ becomes *zero* divided by *zero*. This is not a difficult mathematical problem; as x becomes very small, $sin(x)$ approaches the value of x (see Fig. 11-4).

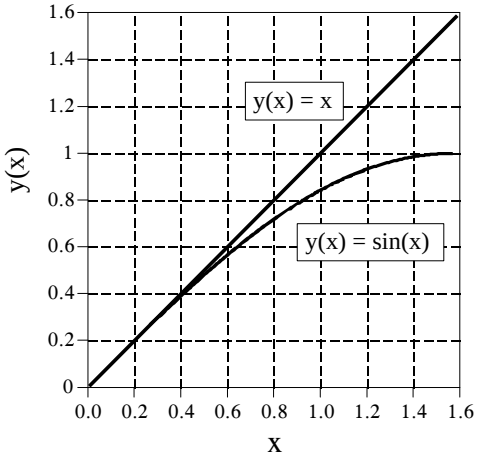

FIGURE 11-4
Comparing x and $sin(x)$. The functions: $y(x) = x$, and $y(x) = sin(x)$ are similar for small values of x, and only differ by about 36% at 1.57 ($\pi/2$). This describes how aliasing distorts the frequency spectrum of the rectangular pulse from a pure sinc function.